

Jose Rene Alvarez

Senior Software Engineer

📍 Sunnyside, Washington
📞 (707) 390-0763
✉ ajoserene70@gmail.com
in linkedin.com/in/jose-rene-alvarez-a029b0353

Summary

Senior Software Engineer with 12+ years of hands-on experience in Node.js, TypeScript, NestJS, React, AWS, and microservices architecture. Adept in Java, Go, Python, and API development, with a keen focus on building event-driven, cloud-native applications. Strong leadership skills in agile teams, driving innovation and delivering efficient, scalable solutions. Known for expertise in test-driven development, optimizing performance, and creating robust technical documentation. Proven track record across real estate, healthcare, and SaaS industries.

Skills

Programming Languages

JavaScript, TypeScript, Java, Golang,
Python, SQL, C#

Software Design & Architecture

Microservices, SOLID Principles, DDD,
OOP, Event-Driven Architecture,
Hexagonal Architecture

Backend Technologies

Node.js, Express.js, NestJS, Sequelize,
TypeORM, Java Spring, Spring
Framework, Spring Boot, GIN, GORM

Frontend Technologies

React, Next.js, Redux, Vue.js, HTML5,
CSS3, SASS, Bootstrap, jQuery

Mobile Development

React Native, Android

Database Management

MySQL, PostgreSQL, MongoDB, Redis,
Amazon Aurora, AWS RDS, AWS
DynamoDB

Experience

Senior Software Engineer

Terrene Homes

01/2021 – Present | Bellevue, WA

- Led the development of a high-performance Terrene home sales platform using Node.js, TypeScript, Java, Next.js, and PostgreSQL, enhancing scalability, user experience, and generating \$4M in monthly revenue.
- Engineered optimized database solutions with PostgreSQL, AWS RDS, TypeORM, PL/pgSQL, and Redis caching, boosting performance and reliability under heavy data loads.
- Designed a scalable backend architecture with Express.js, Spring Framework, microservices, event-driven design, AWS SQS/SNS, and implemented AWS Cognito with OAuth2, OpenID Connect, and JWT for secure, efficient communication and authentication.
- Developed a high-performance Property Listing Service for a Home Sales Platform, leveraging Golang's Goroutines for efficient concurrency, real-time property updates, and high-throughput handling of listing searches and transactions.
- Secured RESTful APIs and gRPC services, ensuring reliable, low-latency communication between microservices while maintaining high performance.
- Managed codebase integrity with Git and GitHub, implemented containerization with Docker, set up CI/CD pipelines using Kubernetes, and utilized Terraform for Infrastructure-as-Code (IaC) to ensure smooth collaboration, version control, and streamlined deployments.
- Leveraged AWS services like Lambda, API Gateway, and S3 for serverless functions, real-time communication, and scalable storage to optimize system performance and costs.
- Developed a high-performance frontend using React, Next.js 14, SSR, Tailwind CSS, and design patterns to optimize load times, SEO, and user experience.
- Designed and developed responsive mobile applications using React Native and the MVVM design pattern, ensuring a seamless, modular user experience across iOS, Android, and Windows platforms.
- Monitored system performance and application health at Terrene Homes, utilizing AWS CloudWatch for log analysis and alerting, while leveraging Grafana for real-time data visualization and dashboarding to ensure platform stability.
- Developed and deployed an AI-driven real estate recommendation and property insights platform for Terrene Homes, utilizing Python, TensorFlow, and an integrated GenAI ChatBot to enhance buyer decision-making, streamline property searches, and improve operational efficiency.
- Implemented comprehensive testing strategies with Jest for unit testing and Selenium for end-to-end testing, ensuring platform reliability and stability.
- Led a team of 3 developers, managing Agile workflows and technical documentation with Jira, driving iterative development, code reviews, and task prioritization to meet project goals.

Full Stack Engineer

Beacon Development Group

09/2015 – 12/2020 | Seattle, WA

- Spearheaded and built a Building Management System at Beacon Development Group, leveraging NestJS, TypeScript, React.js 16, and MySQL to deliver a robust, high-performance application.
- Optimized database queries and performance using TypeORM and PL/pgSQL, reducing query execution time by 70% and improving throughput, while leveraging Amazon Aurora (PostgreSQL) on AWS RDS for scalable data storage and integrating Redis for low-latency caching.

API Development & Integration

RESTful APIs, GraphQL, gRPC,
RabbitMQ, Kafka, AWS SQS/SNS

Cloud Platforms

Amazon Web Service, Lambda, S3, API
Gateway, EC2, Elastic Beanstalk, EKS

DevOps & CI/CD

Docker, Kubernetes, Jenkins, GitHub
Actions, Terraform, Git, GitLab,
BitBucket, CI/CD Pipelines, CircleCI

Monitoring & Logging

CloudWatch, Prometheus, Grafana

AI Frameworks

Tensorflow, PyTorch, scikit-learn

Testing & Quality Assurance

Jest, Cypress, Mocha, Selenium

Methodologies & Soft Skills

Agile, Scrum, Kanban, SDLC,
Collaboration, Mentorship, Leadership

- Enhanced system communication by implementing RESTful APIs, tRPC, and RabbitMQ for seamless inter-service interactions and efficient data processing, while integrating real-time streaming with Kafka to improve service performance and scalability.
- Designed and built a high-performance frontend with TypeScript, React, Next.js, Redux, and React Router, enhancing user experience through Server-Side Rendering (SSR) for faster load times and responsive interactions.
- Established a comprehensive DevOps pipeline using AWS tools, such as Lambda, S3, EC2, EKS, Elastic Beanstalk, and API Gateway, along with Docker, Kubernetes, Jenkins, Terraform, CircleCI, and GitHub Actions, to automate deployments and scale infrastructure seamlessly.
- Utilized AWS CloudWatch, Grafana, and Prometheus to monitor and analyze system performance, ensuring consistent uptime and quick identification of issues.
- Worked within an agile development team, managing tasks through Scrum and creating comprehensive internal documentation via ClickUp.

Full Stack Developer

Evershine Med

11/2011 – 09/2015 | Austin, TX

- Played a key role in developing and optimizing a scalable Patient Management System using Node.js, Java, React, and MongoDB, ensuring the solution's long-term maintainability, flexibility, and ability to meet evolving business needs.
- Enhanced backend performance by implementing Node.js, Express.js, Java Spring, Sequelize, and Hibernate to build high-performance APIs, ensuring fast and reliable data access for high-demand operations.
- Improved database schema design and optimized query performance in MongoDB using aggregation pipelines, while incorporating Redis to significantly reduce response times and enhance data retrieval efficiency, even under heavy traffic loads.
- Developed serverless functions using AWS Lambda to streamline processing, while leveraging AWS S3 for secure, scalable file storage, and API Gateway to implement efficient RESTful APIs and GraphQL for seamless data exchange.
- Elevated frontend performance by optimizing React, Redux, Redux Saga, and JavaScript, incorporating techniques like lazy loading, advanced state management, and enhancing user experience through faster page loads and smoother interactions.
- Led quality assurance initiatives by writing and executing comprehensive unit and integration tests using Jest, JUnit, and Cypress, adhering to Agile methodologies, and leveraging Atlassian Jira to manage tasks, track progress, and resolve bugs effectively.

Education

Bachelor's Degree in Computer Science

Arizona State University

- 09/2004 – 05/2009 | Tempe, AZ